



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

JUNE 2006

TASK #3

YEAR 12

Mathematics

General Instructions:

- Reading time—5 minutes.
- Working time—2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections:
Section A(Questions 1 and 2),
Section B(Questions 3 and 4),
Section C(Questions 5 and 6),

Total marks—100 Marks

- Attempt questions 1–6.
- All questions are NOT of equal value.

Examiner: Mr R. Boros

This is an assessment task only and does not necessarily reflect
the content or format of the Higher School Certificate.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln (x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln (x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A

Marks

Question 1 (20 marks)

(a) Differentiate with respect to x :

(i) $x^4 - \sqrt{x} + 1$

1

(ii) $x \tan x$

1

(iii) $\cos^2 3x$

1

(iv) xe^{2x}

1

(v) $(x + 2) \ln(x + 2)$

2

(b) Find a primitive of:

(i) $2x^2 + 4x - 1$

1

(ii) $\frac{2}{2x + 3}$

1

(iii) $3 \sin 2x$

1

(iv) $\sec^2(3x + 1)$

1

(c) If $\frac{dy}{dx} = 6x - 1$ and the function passes through $(1, 2\frac{1}{2})$,
find y as a function of x .

2

(d) Consider the curve $y = x(x^2 - 12x + 45)$.

(i) Find the coördinates of all the stationary points and determine their nature.

3

(ii) Find the coördinates of the points of inflexion.

2

(iii) Sketch the curve, showing all important features.

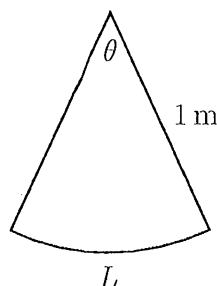
1

(iv) Find the equation (written in general form) of the tangent to the curve at the origin.

2

Question 2 (18 marks)

(a)



A pendulum of length 1 m takes one second to swing from one side to the other. It travels at 180 cm/min.

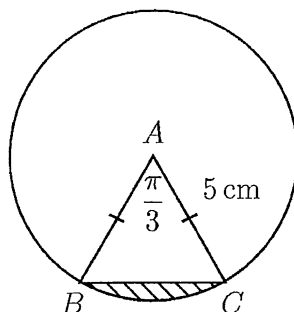
(i) Find the distance L through which it swings.

2

(ii) Calculate the angle θ (in degrees and minutes) through which it swings each second.

2

(b)



Consider the above diagram, not drawn to scale. Give answers correct to the nearest cm^2 .

(i) Find the area of the sector ABC .

1

(ii) Find the area of the shaded segment.

2

(c) State the largest possible domain of the function $y = \ln(1 - 4x^2)$.

1

(d) (i) Sketch the parabola $y = 6x - 3x^2$, showing the x -intercepts and the vertex.

2

(ii) Find the area enclosed by the parabola in (d)(i) and the x -axis from $x = -1$ to $x = 1$.

3

(e) Find, correct to 2 decimal places,

(i) $\int_{-1}^0 e^{5-2x} dx,$

2

(ii) $\int_0^2 \tan^2\left(\frac{x}{2}\right) dx.$

2

(f) Solve, correct to 3 decimal places, the equation $e^{2x} = \ln 1994$.

1

Section B

(Use a separate writing booklet.)

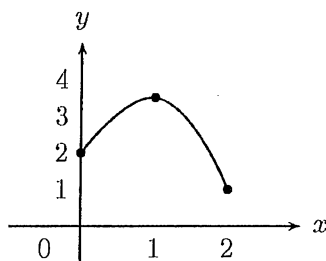
Marks

Question 3 (15 marks)

- (a) (i) Sketch the curves $y = \sin x$ and $y = 1 + \sin x$ on the same axes for $0 \leq x \leq \pi$. 2

- (ii) The region between the curves in 3(a)(i) is rotated about the x -axis, $0 \leq x \leq \pi$. Find the volume of the solid of revolution in exact form. 3

- (b) A curve $y = f(x)$ is known to pass through the points $(0, 2)$, $(1, 3\frac{1}{2})$, $(2, 1)$. 2



The region bounded by the curve and the x -axis from $x = 0$ to $x = 2$ is rotated about the x -axis. Use Simpson's Rule to approximate the volume of the solid of revolution.

- (c) Find the equation (written in general form) of the tangent to the curve $y = e^{x^2}$ at the point $(1, e)$ on it. 2

- (d) A pool is being drained and the number of litres of water, L , in the pool at time t minutes is given by the equation

$$L = 120(40 - t)^2.$$

- (i) At what rate is the water draining out of the pool when $t = 6$ minutes? 2

- (ii) How long will it take for the pool to be completely empty? 1

- (e) Show that the curve $y = x^3 + 3x + 1$ is increasing for all values of x . 2

- (f) Sketch the graph of the function which has all these features: 1

$$\begin{aligned} f(2) &= 0, & f'(2) &= 0, \\ f'(x) &< 0 & \text{for all } x < 2, \\ f'(x) &> 0 & \text{for all } x > 2. \end{aligned}$$

Question 4 (17 marks)

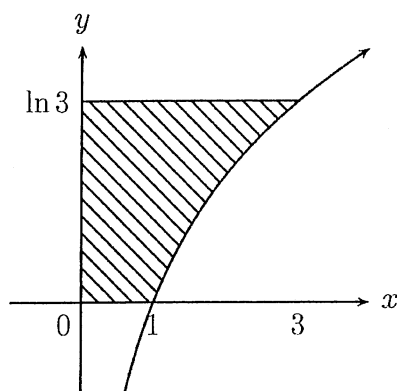
- (a) A spherical bubble is being inflated so that the rate of change, R , of the volume, V , in cm^3/s , at any instant of time, t , in seconds, is given by

4

$$R = \frac{4t}{t^2 + 1}, \quad t \geq 0.$$

Initially the volume of the bubble was 40 cm^3 . Given that the volume of a sphere with radius r is $\frac{4}{3}\pi r^3$, find the radius of the bubble after 100 seconds, correct to 2 significant figures.

(b)



The diagram shows the area bounded by the graph $y = \ln x$, the coördinate axes, and the line $y = \ln 3$.

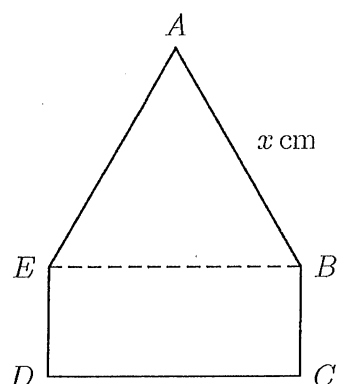
- (i) Find the size of the shaded area.

2

- (ii) Hence or otherwise find the exact value of $\int_1^3 \ln x \, dx$.

2

(c)



$ABCDE$ is a pentagon of fixed perimeter P cm.
In the figure, triangle ABE is equilateral and
 $BCDE$ is a rectangle.
The length of AB is x cm.

(i) Show that the length of BC is $\frac{P-3x}{2}$ cm.

1

(ii) Show that the area of the pentagon is given by

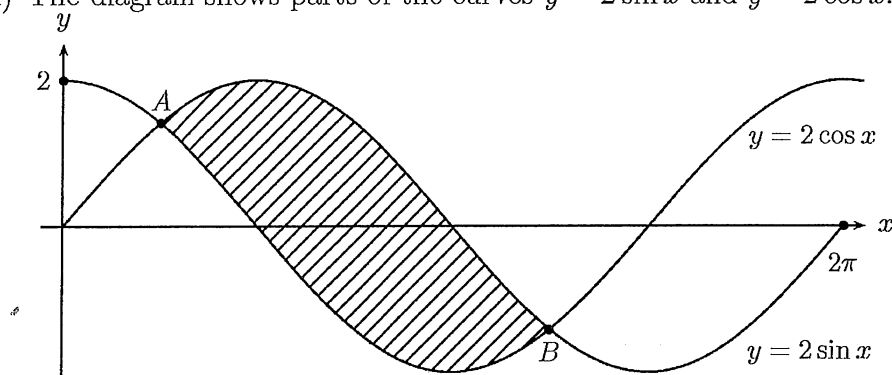
2

$$A = \frac{1}{4} [2Px - (6 - \sqrt{3})x^2] \text{ cm}^2.$$

(iii) Find the value of $\frac{P}{x}$ for which the area of the pentagon is a maximum.

2

(d) The diagram shows parts of the curves $y = 2 \sin x$ and $y = 2 \cos x$.



(i) Find the points A and B .

2

(ii) Hence calculate the shaded area (leaving your answer in exact form).

2

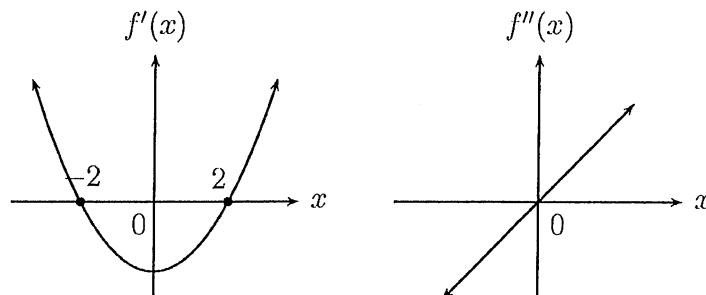
Section C

(Use a separate writing booklet.)

Marks

Question 5 (15 marks)

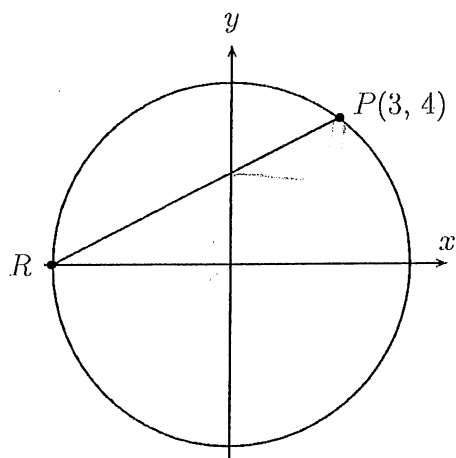
- (a) The graphs below show the first and second derivatives of a curve $y = f(x)$.



- (i) For which values of x is the function
(α) increasing, 2
(β) concave down?
- (ii) Give the x -coördinate for the maximum turning point. 2
- (b) (i) Sketch the graph of $y = 3 \cos 2x$ in the range $0 \leq x \leq 2\pi$. 2
- (ii) Using your graph of 5(b)(i), find how many solutions there are to the equation $\cos 2x = \frac{1}{3}$ in the range $0 \leq x \leq 2\pi$. 1
- (c) The number of vehicles N , at time t , in the City of Sydney has been graphed over a period of 10 years. It was found that over the whole period
$$\frac{dN}{dt} > 0 \text{ while } \frac{d^2N}{dt^2} < 0.$$

(i) What does this tell you about the number of vehicles in Sydney over this time? 1
- (ii) What can be said about the rate of change in the number of cars in Sydney over this period of time? 1
- (iii) Make a neat sketch of N against t . 1

(d)



$P(3, 4)$ is a point on the circle $x^2 + y^2 = 25$. Find the length of the minor arc PR correct to 3 significant figures.

3

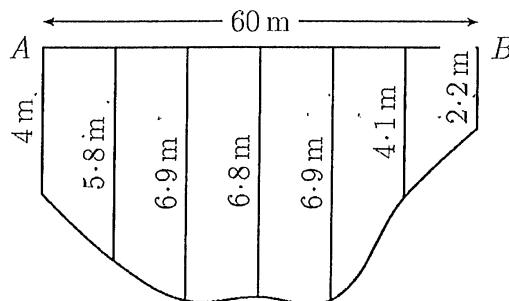
(e) Given the function $f(x) = xe^{-x^2}$, show that this is an odd function.

2

Question 6 (15 marks)

Marks

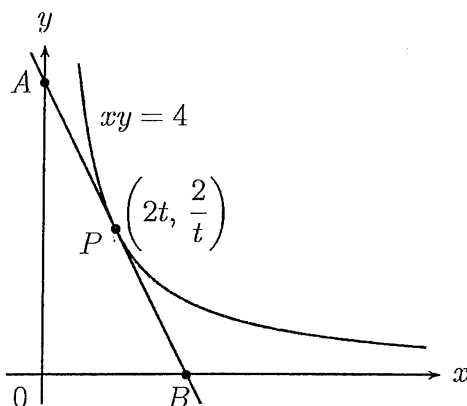
(a)



2

The diagram represents the cross section of a stream which is 60 m wide. The depth of the stream was measured at 10 m intervals from bank A to bank B. Use the trapezoidal rule to approximate the area of the cross-section of the river.

(b)



In the diagram, $P(2t, \frac{2}{t})$ is a variable point on the branch of the hyperbola $xy = 4$ in the first quadrant. The tangent at P meets the y -axis at A and the x -axis at B .

(i) Show that the tangent at P is $t^2y = 4t - x$.

3

(ii) Let the square of the length of AB , i.e. $(AB)^2$, be denoted by W . Find the minimum value of W .

4

(c) A patient sick in hospital is being treated with a new medicine to fight bacteria present in the bloodstream. The number of bacteria present, N , t hours after the medicine is administered is given by the equation $N = Be^{-0.04t}$, where B is a constant.

(i) Show that the rate at which the number of bacteria decreases is proportional to the number of bacteria present. 1

(ii) After h hours the number of bacteria has halved. Find the value of h (to one decimal place). 2

(iii) After 36 hours the number of bacteria is estimated at 5×10^4 . Find the value of B (to the nearest whole number). 1

(iv) The patient is discharged when the number of bacteria reduces to 10 000. When can the patient leave hospital? (Use the value of B found in part (iii) and answer to the nearest whole hour) 2

End of Paper

SECTION A, 2006 HSC TASK 3, MATHEMATICS

QUESTION ONE

$$\begin{aligned} \text{a) i) } \frac{d}{dx}(x^4 - \sqrt{x} + 1) \\ = 4x^3 - \frac{1}{2\sqrt{x}}. \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{d}{dx}(x \tan x) \\ = \tan x + x \sec^2 x. \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{d}{dx}(\cos^2 3x) \\ = -6 \cos 3x \sin 3x. \end{aligned}$$

$$\begin{aligned} \text{iv) } \frac{d}{dx}(x e^{2x}) \\ = e^{2x} + 2x e^{2x} \\ = e^{2x}(1 + 2x). \end{aligned}$$

$$\begin{aligned} \text{v) } \frac{d}{dx}((x+2) \ln(x+2)) \\ = \ln(x+2) + x+2 \times \frac{1}{x+2} \\ = \ln(x+2) + 1. \end{aligned}$$

$$\begin{aligned} \text{b) i) } \int 2x^2 + 4x - 1 \cdot dx \\ = \frac{2}{3}x^3 + 2x^2 - x + C. \end{aligned}$$

$$\text{ii) } \int \frac{2}{(2x+3)} \cdot dx = \ln(2x+3) + C$$

$$\begin{aligned} \text{iii) } \int 3 \sin 2x \cdot dx &= 3 \int \sin 2x \cdot dx \\ &= -\frac{3}{2} \cos 2x + C \end{aligned}$$

$$\begin{aligned} \text{iv) } \int \sec^2(3x+1) \cdot dx \\ = \frac{1}{3} \tan(3x+1) + C. \end{aligned}$$

$$\text{c) } \frac{dy}{dx} = 6x - 1 \text{ thro' } (1, 2\frac{1}{2})$$

$$\begin{aligned} y &= \int 6x - 1 \cdot dx \\ &= 3x^2 - x + C. \end{aligned}$$

$$\text{When } y = 2\frac{1}{2}, x = 1.$$

$$\therefore 2\frac{1}{2} = 3(1) - 1 + C$$

$$\therefore C = \frac{1}{2}.$$

SECTION A, 2006, HSC TASK #3 MATHEMATICS

QUESTION ONE (CONT)

d) Stationary points
when $\frac{dy}{dx} = 0$

$$y = x^3 - 12x^2 + 45x.$$

$$\frac{dy}{dx} = 3x^2 - 24x + 45.$$

$$\therefore 3x^2 - 24x + 45 = 0.$$

$$3(x-5)(x-3) = 0.$$

$$x=5 \quad \text{or} \quad x=3.$$

$$y=50$$

$$y=54.$$

Turning points occur at.

$$A(5,50) \text{ \& } B(3,54)$$

NATURE

$$\underline{A} \quad \frac{d^2y}{dx^2} = 6x - 24$$

$$= 6(5) - 24$$

$$> 0$$

$\therefore$ minimum at $A(5,50)$

$$B // \quad \frac{d^2y}{dx^2} = 6(3) - 24$$

$$< 0$$

$\therefore$ maximum at $B(3,54)$

i) P.O.I when $\frac{dy}{dx} = 0$

$$\therefore 6x - 24 = 0$$

$$x = 4$$

$$y = 52.$$

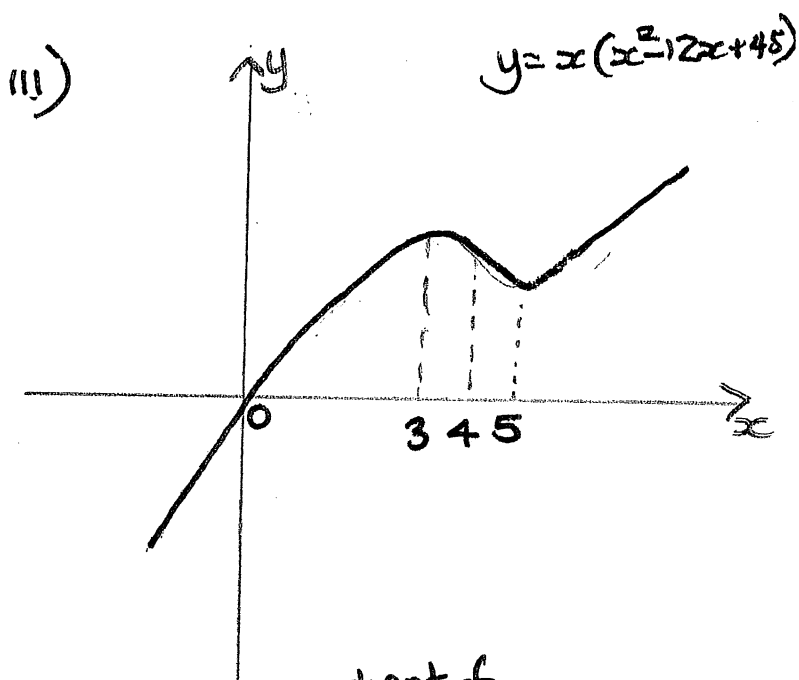
$\therefore$ possible P.O.I @ $(4,52)$

CONCAVITY

$$x > 4 \quad \frac{d^2y}{dx^2} > 0 \quad \text{concave up.}$$

$$x < 4 \quad \frac{d^2y}{dx^2} < 0 \quad \text{concave down.}$$

Since there is a change in concavity there is a P.O.I at $(4,52)$.



iv) at origin, gradient of tangent given by $\frac{dy}{dx}$.

$$\frac{dy}{dx} = 3x^2 - 24x + 45.$$

when $x=0$.

$$\frac{dy}{dx} = 45$$

$\therefore$ pt gradient formula gives $y-0 = 45(x-0)$

$$45x - y = 0$$

is eqn of tangent at $(0,0)$

SECTION A, 2006, HSC TASK #3 MATHEMATICS.

QUESTION TWO.

a) i) Travels @ 180 cm/min.

∴ travels 3 cm/sec.

∴ length of swing = 3 cm.

ii) $l = r \theta$ $l = 3 \text{ cm}$ $r = 100 \text{ cm}$.

$$3 \text{ cm} = 100 \theta$$

$$\theta = \frac{3}{100}^\circ$$

$$\underline{\underline{0.03^\circ = 1^\circ 43' 8''}}$$

$$\begin{aligned} \text{b) i) Area of Sector ABC} &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \times 25 \times \frac{\pi}{3} \\ &= \underline{\underline{13.09 \text{ cm}^2}}. \end{aligned}$$

$$\text{ii) AREA OF SEGMENT} = \text{Area of sector} - \text{Area } \triangle ABC.$$

$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2} bc \sin A. \\ &= \frac{1}{2} (25) \sin \frac{\pi}{3}. \\ &= 10.83 \text{ cm}^2. \end{aligned}$$

$$\begin{aligned} \text{AREA OF SEGMENT} &= 13.09 - 10.83. \\ &= \underline{\underline{2.26 \text{ cm}^2}}. \end{aligned}$$

QUESTION TWO (CONT)

c) $y = \ln(1 - 4x^2)$

$$1 - 4x^2 > 0$$

$$x^2 < \frac{1}{4}$$

$$\underline{\underline{-\frac{1}{2} < x < \frac{1}{2}}}$$

d) i) $y = 6x - 3x^2$

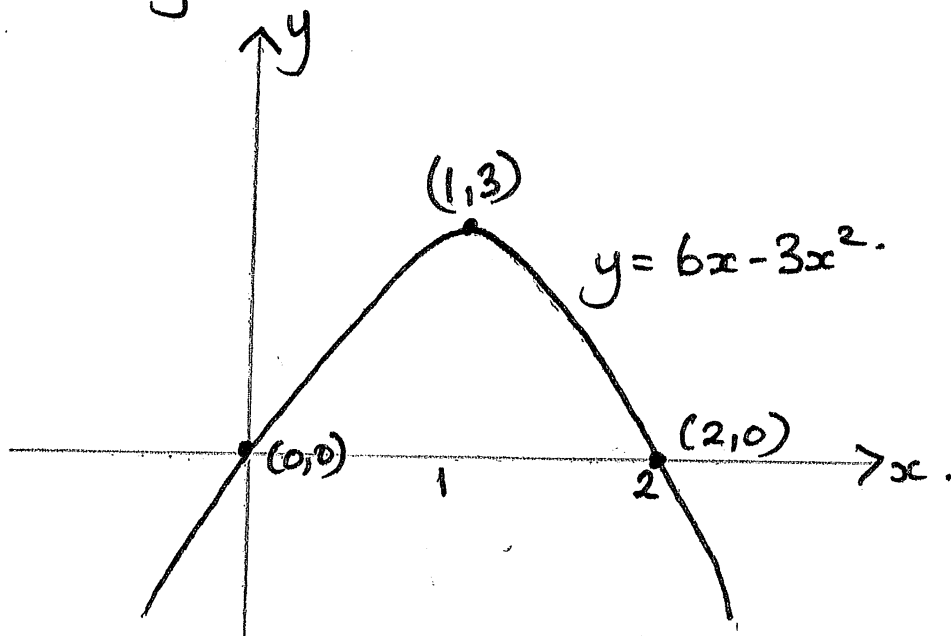
∴ $y = 3x(2 - x)$

x intercepts at $x=0$ & $x=2$.
 $y=0$ $y=0$

x. Vertex = $\frac{-b}{2a}$

$$x = 1$$

$$y = 3$$



QUESTION TWO (CONT)

$$\begin{aligned}
 \text{d) ii) Area} &= \left| \int_{-1}^0 y \cdot dx \right| + \int_0^1 y \cdot dx. \\
 &= \left| [3x^2 - x^3]_{-1}^0 \right| + [3x^2 - x^3]_0^1 \\
 &= |-2| + 4 \\
 &= \underline{\underline{6 \text{ units}^2}}.
 \end{aligned}$$

$$\begin{aligned}
 \text{e) i) } \int_{-1}^0 e^{5-2x} \cdot dx &= \left[-\frac{1}{2} e^{5-2x} \right]_{-1}^0 \\
 &= -\frac{1}{2} e^5 + \frac{1}{2} e^7 \\
 &= \underline{\underline{474.11 \text{ units}^2. (2 \text{ dp})}}
 \end{aligned}$$

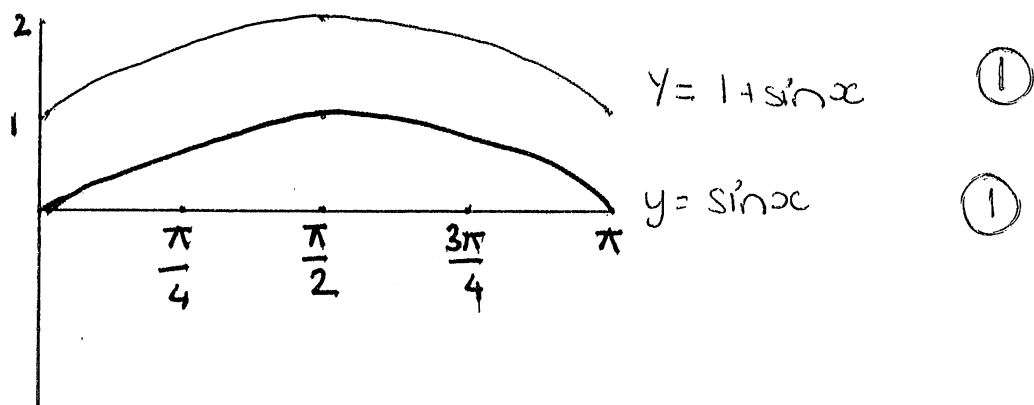
$$\begin{aligned}
 \text{ii) } \int_0^2 \tan^2\left(\frac{x}{2}\right) \cdot dx &= \int_0^2 \sec^2\left(\frac{x}{2}\right) - 1 \cdot dx. \\
 &= \left[2 \tan\left(\frac{x}{2}\right) - x \right]_0^2 \\
 &= (2 \tan 1 - 2) - (2 \tan 0 - 0) \\
 &= 1.1148 \\
 &= \underline{\underline{1.11 (2 \text{ dp})}}
 \end{aligned}$$

$$\begin{aligned}
 \text{f) } e^{2x} &= \ln 1994. \\
 \ln e^{2x} &= \ln(\ln 1994). \\
 2x &= \ln(\ln 1994) \\
 x &= \underline{\underline{1.014 (3 \text{ dp})}}.
 \end{aligned}$$

Section B

Question 3:

a) i)



2

b) ii)

$$V = \pi \int_0^{\pi} (1 + \sin x)^2 dx - \pi \int_0^{\pi} (\sin x)^2 dx \quad \text{①/2}$$

$$= \pi \int_0^{\pi} [(1 + \sin x)^2 - (\sin x)^2] dx$$

$$= \pi \int_0^{\pi} [1 + 2\sin x + \sin^2 x - \sin^2 x] dx$$

$$= \pi \int_0^{\pi} [1 + 2\sin x] dx$$

$$= \pi [x - 2\cos x]_0^{\pi} \quad \text{①}$$

$$= \pi [(\pi - 2\cos \pi) - (0 - 2\cos 0)] \quad \text{①/2}$$

$$= \pi [\pi + 2 - 0 + 2]$$

$$= \pi [\pi + 4] \quad \text{①}$$

3

1 mark for

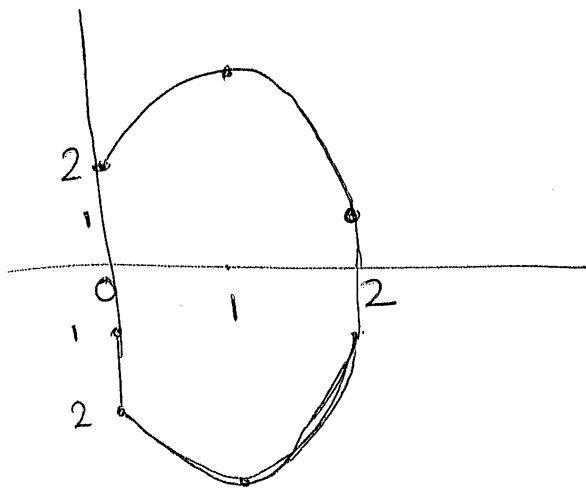
$$y^2 = \sin(1 + \sin x - \sin x)$$

$$= 12$$

$$= 1$$

Question 5

c)



x	0	1	2
$f(x)$	2	$3\frac{1}{2}$	1
$f(x)^2$	4	$12\frac{1}{4}$	1

$$V = \pi \int_0^2 f(x)^2 dx.$$

$$\begin{aligned}\text{Simpsons Rule} &= \frac{1}{3} [4 + 1 + 4(12\frac{1}{4})] \\ &= 18. \quad \textcircled{1}\end{aligned}$$

1 for $\pi \times \text{SR for } f(x)$

$$V = \pi \times 18$$

$$= 18\pi \text{ ①}$$

1 for SR
 $f(x)^2$

$\frac{1}{2}$

$\frac{1}{2}$ only for
simpsons rule $f(x)$
 $\frac{2}{3} [2 + 4(3.5) + 1]$

Question 3.

$$2) \quad y = e^{x^2}$$

point (1, e)

$$\frac{dy}{dx} = e^{x^2} \times 2x$$

$$= 2xe^{x^2} \quad (1)$$

when $x = 1$

$$\frac{dy}{dx} = 2 \cdot 1 \cdot e^{1^2}$$
$$= 2e \quad (1/2)$$

$$\therefore T: \quad \frac{y-e}{x-1} = 2e$$

$$y-e = 2e(x-1)$$

$$y-e = 2ex - 2e$$

$$2ex - y - e = 0 \quad (1/2)$$

-1/2 for
not gen form.

2

$$3) \quad L = 120(40-t)^2$$

$$1) \quad \frac{dL}{dt} = 240(40-t) \times -1$$
$$= -240(40-t) \quad (1)$$

1/2 for $\frac{dy}{dt}$
if whole answer
wrong.

when $t = 6$

$$\frac{dL}{dt} = -240(40-6)$$
$$= -240(34)$$

$$= -8160 \text{ Lm}^{-1}$$

$\therefore$ draining out at $8160 \text{ Lm}^{-1} \quad (1)$

2

d) ii) $L=0$ $t=?$

$$0 = 120(40 - t)^2$$

$$(40 - t)^2 = 0$$

$$40 - t = 0$$

$$t = 40 \text{ (1)}$$

$\therefore$ The pool will be empty after 40 minutes.
1.

e). $y = x^3 + 3x + 1.$

$$y' = 3x^2 + 3 \text{ (1/2)}$$

for increasing curve $y' > 0.$

$$3x^2 + 3 > 0 \text{ (1/2)}$$

~~$$\begin{aligned}
 3x^2 &> -3 \\
 x^2 &> -1 \\
 x &> \pm \sqrt{-1}
 \end{aligned}$$~~

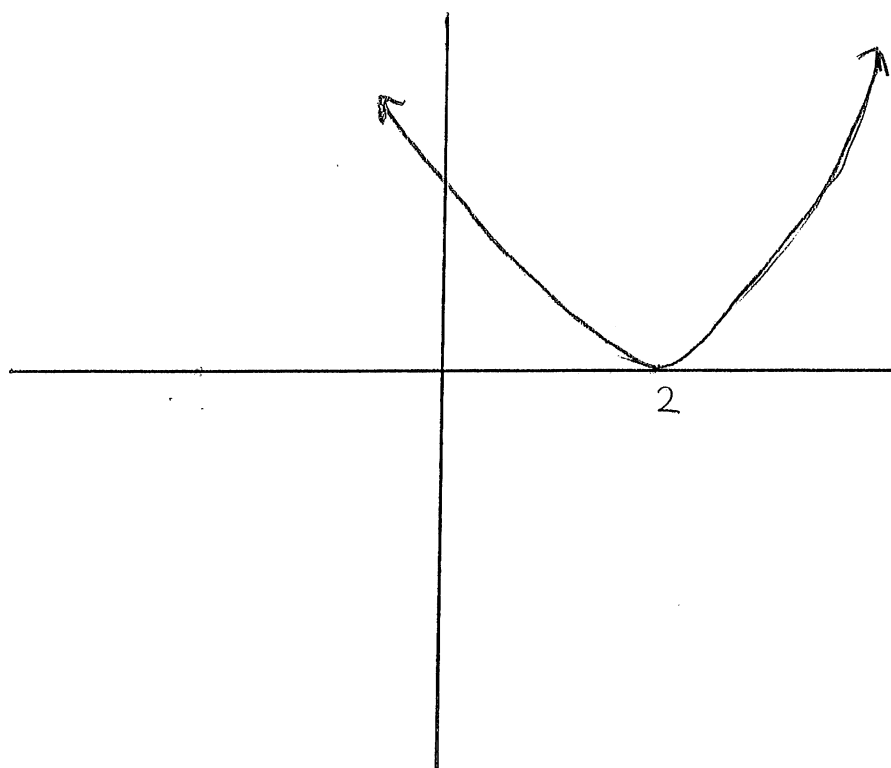
x^2 will always be positive

$\therefore 3x^2 + 3$ will always be positive (1)

$\therefore f'(x) > 0$ for all x

$\therefore f(x)$ is increasing for all $x.$

f)



1

Question 4

$$1) R = \frac{4t}{t^2+1} = \frac{dv}{dt}$$

$$V = \int \frac{4t}{t^2+1} dt$$

$$= 2 \int \frac{2t}{t^2+1} dt$$

$$= 2 \ln(t^2+1) + C. \textcircled{1}$$

when $t=0$, $V=40$

$$40 = 2 \ln(0^2+1) + C$$

$$40 = 0 + C$$

$$C = 40$$

$$\therefore V = 2 \ln(t^2+1) + 40 \textcircled{1}$$

when $t=100$

$$V = 2 \ln(100^2+1) + 40$$

$$= 2 \ln(10001) + 40$$

$$= 58.42088073 \textcircled{1}$$

$$V = \frac{4}{3} \pi r^3$$

$$58.42088073 = \frac{4}{3} \pi r^3$$

$$r^3 = 58.42088073 \times \frac{3}{4\pi}$$

$$r = 2.407094627$$

$$= 2.4 \text{ cm (2 sig fig)} \textcircled{1}$$

4

Question 4

b) i) $y = \ln x$

$$\therefore x = e^y \quad (1)$$

$$A = \int_0^{\ln 3} e^y$$

$$= [e^y]_0^{\ln 3} \quad (1/2)$$

$$= 3 - 1$$

$$= 2 \quad (1/2)$$

/2

ii) $\int_1^3 \ln x \, dx$

$$A = \text{Area of rectangle} - \text{shaded region.} \quad (1/2)$$

$$= 3 \times \ln 3 \quad (1/2) - 2$$

$$= 3 \ln 3 - 2. \quad (1)$$

/2

Question 4

$$\begin{aligned} \therefore i) P &= AE + AB + BC + CD + DE \\ &= x + x + BC + x + DE \\ &= 3x + BC + DE \end{aligned}$$

$$P - 3x = BC + DE$$

AS BCDE is a rectangle $BC = DE \Rightarrow$ opposite sides of rectangle BCDE.

$$\therefore P - 3x = BC + BC$$

$$P - 3x = 2BC$$

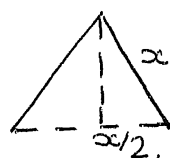
$$\therefore BC = \frac{P - 3x}{2} \quad (1)$$

$\therefore$ i) $A =$ area of triangle + area of rectangle.

$$\text{Area of triangle} = \frac{1}{2} b h$$

$$= \frac{1}{2} \times x \times \frac{\sqrt{3}x}{2}$$

$$= \frac{\sqrt{3}x^2}{4} \quad (1/2)$$



$$\begin{aligned} h^2 &= x^2 - \left(\frac{x}{2}\right)^2 \\ &= x^2 - \frac{x^2}{4} \end{aligned}$$

$$h^2 = \frac{3x^2}{4}$$

$$h = \frac{\sqrt{3}x}{2} \quad (1/2)$$

$$\text{Area of rectangle} = \frac{P - 3x}{2} \times x$$

$$= \frac{Px - 3x^2}{2} \quad (1/2)$$

$$A = \frac{\sqrt{3}x^2}{4} + \frac{Px - 3x^2}{2}$$

$$A = \frac{\sqrt{3}x^2 + 2Px - 6x^2}{4}$$

$$A = \frac{1}{4} [\sqrt{3}x^2 + 2Px - 6x^2]$$

$$A = \frac{1}{4} [2Px - (6 - \sqrt{3})x^2] \text{ cm}^2 \quad (1/2)$$

/2

$$\text{ii) } A = \frac{Px}{2} - \frac{6x^2}{4} + \frac{\sqrt{3}x^2}{4}$$

$$\frac{dA}{dP} = \frac{P}{2} - \frac{12x}{4} + \frac{2\sqrt{3}x}{4}$$

let $dA/dP = 0$ to find stationary point.

$$\therefore 0 = \frac{P}{2} - \frac{12x}{4} + \frac{2\sqrt{3}x}{4} \quad (1/2)$$

$$0 = 2P - 12x + 2\sqrt{3}x$$

$$12x - 2\sqrt{3}x = 2P$$

$$6x - \sqrt{3}x = P$$

$$(6 - \sqrt{3})x = P$$

$$6 - \sqrt{3} = P/x. \quad (1)$$

$$\frac{d^2A}{dx^2} = -3 + \frac{\sqrt{3}}{2} \quad (1/2)$$

$$= -2.133974596$$

$\therefore 6 - \sqrt{3}$ is a max. turning point.

$\therefore$ when $P/x = 6 - \sqrt{3}$ the area of the pentagon is a maximum.

/2

Question 4

i) $y = 2 \sin x, \quad y = 2 \cos x$

$$\therefore 2 \sin x = 2 \cos x$$

$$\therefore \sin x = \cos x \quad (1)$$

$$\sin x = \cos x \quad \text{between} \quad 0 \leq x \leq 2\pi \quad \text{at}$$

$$\pi/4 \text{ \& } 5\pi/4 \quad (1)$$

/2

ii) $A = \int_{\pi/4}^{5\pi/4} [2 \sin x - 2 \cos x] dx$

$$= 2 \int_{\pi/4}^{5\pi/4} [\sin x - \cos x] dx$$

$$= 2 \left[-\cos x - \sin x \right]_{\pi/4}^{5\pi/4} \quad (1)$$

$$= 2 \left[(-\cos 5\pi/4 - \sin 5\pi/4) - (-\cos \pi/4 - \sin \pi/4) \right]$$

$$= 2 \left[\left(\frac{1}{\sqrt{2}} - - \frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \right]$$

$$= 2 \left[\frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} \right]$$

$$= 4\sqrt{2} \cup^2 \quad (1)$$

/2

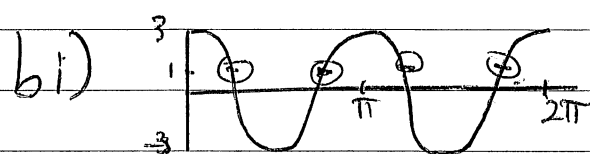
SECTION C

QUESTION 5.

a) $x < -2, x > 2.$

$\beta) x < 0$

ii) $x = -2.$

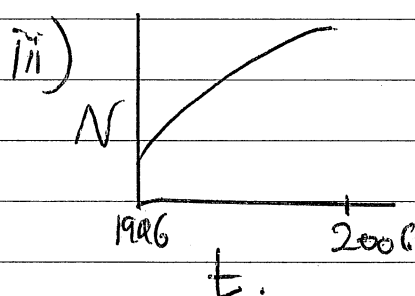


Amplitude = 3
Period = π

ii) 4 solutions

c) i) The number of vehicles is increasing.

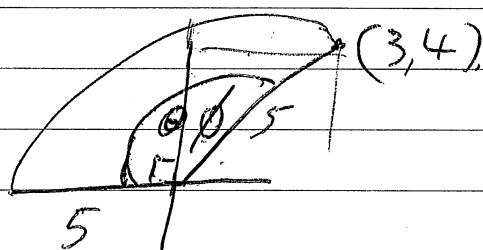
ii) The rate of increase is decreasing.



d) $\tan \phi = \left(\frac{3}{4}\right)$

$\phi = \tan^{-1}\left(\frac{3}{4}\right)$

$\theta = \phi + \frac{\pi}{2}$



$L = r\theta$

$= 5 \times \left(\tan^{-1}\left(\frac{3}{4}\right) + \frac{\pi}{2}\right)$

$= 11.1$

$$\begin{aligned}
 e) f(-a) &= -ae^{-(-a)^2} \\
 &= -ae^{-a^2} \\
 &= -f(a).
 \end{aligned}$$

QUESTION 6.

$$\begin{aligned}
 a) A &\doteq \frac{10}{2} (4 + 2.2 + 2(5.8 + 6.9 + 6.8 + 9.6 + 4.1)) \\
 &= 336 \text{ m}^2.
 \end{aligned}$$

$$b) y = 4x^{-1}$$

$$\frac{dy}{dx} = -4x^{-2}$$

$$\text{At } x = 2t$$

$$m = -\frac{1}{t^2}$$

$$\therefore y - \frac{2}{t} = -\frac{1}{t^2} (x - 2t)$$

$$t^2 y = 4t - x.$$

$$ii) A \text{ when } x = 0.$$

$$y = \frac{4}{t}$$

$$B \text{ when } y = 0$$

$$x = 4t.$$

By Pythagoras.

$$W = (4t)^2 + \left(\frac{4}{t}\right)^2$$

$$\therefore W = 16t^2 + \frac{16}{t^2}$$

$$\frac{dW}{dt} = 32t - 32t^{-3}$$

$$\frac{d^2W}{dt^2} = 32 + 96t^{-4}$$

Stat pts when $\frac{dW}{dt} = 0$.

$$0 = 32t - 32t^{-3}$$

$$t^4 - 1 = 0$$

$$t = 1, -1$$

$$\frac{d^2W}{dt^2} = 32 + 96(1)^{-4}$$

$$= 128 > 0 \quad \text{minima}$$

$$\frac{d^2W}{dt^2} = 32 + 96(-1)^{-4}$$

$$= 128 > 0 \quad \text{minima}$$

$\therefore$ minimum at $t = 1, -1$.

$$\text{i.e. } W = 32.$$

$$c) \frac{dN}{dt} = -0.04 B e^{-0.04t}$$

$$= -0.04 N.$$

$$ii) \frac{1}{2} B = B e^{-0.04h}$$

$$h = \frac{\ln \frac{1}{2}}{-0.04}$$

$$= 17.3 \text{ hours.}$$

$$iii) 5 \times 10^4 = B e^{-0.04(36)}$$

$$B = 211035 \text{ bacteria.}$$

$$iv) 10000 = 211035 e^{-0.04t}$$

$$t = \frac{\ln\left(\frac{10000}{211035}\right)}{-0.04}$$

$$= 76 \text{ hours.}$$